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In silico prediction, molecular docking and binding studies of acetaminophen and dexamethasone to *Enterococcus faecalis* diaminopimelate epimerase

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Abstract

Enterococcus faecalis (*E. faecalis*) is a Gram-positive coccoid, non-sporulating, facultative anaerobic, multidrug resistance bacterium responsible for almost 65% to 80% of all enterococcal nosocomial infections. It usually causes infective endocarditis, urinary tract and surgical wound infections. The increase in *E. faecalis* resistance to conventionally available antibiotic has rekindled intense interest in developing useful antibacterial drugs. In *E. faecalis*, diaminopimelate epimerase (DapF) is involved in the lysine biosynthetic pathway. The product of this pathway is precursors of peptidoglycan synthesis, which is a component of bacterial cell wall. Also, because mammals lack this enzyme, consequently *E. faecalis* diaminopimelate epimerase (*Ef*DapF) represents a potential target for developing novel class of antibiotics. In this regard, we have successfully cloned, overexpressed the gene encoding DapF in BL-21(DE3) and purified with Ni-NTA Agarose resin. In addition to this, binding studies were performed using fluorescence spectroscopy in order to confirm the bindings of the identified lead compounds (acetaminophen and dexamethasone) with *Ef*DapF. Docking studies revealed that acetaminophen found to make hydrogen bonds with Asn72 and Asn13 while dexamethasone interacted by forming hydrogen bonds with Asn205 and Glu223. Thus, biochemical studies indicated acetaminophen and dexamethasone, as potential inhibitors of *Ef*DapF and eventually can reduce the catalytic activity of *Ef*DapF.

KEYWORDS
docking, *E. faecalis* diaminopimelate epimerase, in silico modelling, potential *Ef*DapF inhibitors, spectrofluorimetric binding studies

1 | INTRODUCTION

The incidence of infections caused by Gram-positive multidrug resistant (MDR) bacterium, *E. faecalis* is very common in hospital environment. It usually causes infections ranging from endocarditis to urinary tract.[1,2] The increase in *E. faecalis* resistance to conventionally available antibiotic has rekindled intense interest in developing useful antibacterial drugs.[3] In *E. faecalis*, PLP-independent amino acid racemases is the diaminopimelate epimerase (DapF) that is involved in the biosynthesis of meso-DAP and lysine in the lysine biosynthetic pathway.[4-6] Meso-DAP and lysine, the products of this pathway are important precursors for the synthesis of peptidoglycan, housekeeping proteins, virulence factors and thus survival of bacteria. Therefore, the design of tight inhibitors of *Ef*DapF may lead to new class of antibacterial drugs.[7] Also, because humans lack this enzyme, consequently *Ef*DapF represents a potential target for the developing novel class of antibiotics.[8,9] Till now, the structure of Dap has been solved from bacterial species including *Haemophilus influenza*,[10,11] *Escherichia coli*,[6,12] *Corynebacterium glutmicum*,[13] *Mycobacterium tuberculosis*,[14] *Arabidopsis thaliana*[15] and are available in protein data bank (PDB). Several other structures are also available in the PDB, but are to date otherwise unpublished, including *Acinetobacter baumannii* (PDB: 5HA4), *Bacillus anthracis* (PDB: 2OTN). Since the first structural study of DAP epimerase was reported in 1998 from *Haemophilus influenzae*,[10] structural and functional studies of DAP epimerase from different organisms have reported that the structure consists of two structurally similar α/β domain and each domain provides one of the two key active site cysteine residues. These cysteine residues are located in the inter-domain cleft, important for catalysis.[6,10-15] In addition, DAP epimerase utilizes a pair of cysteine residues for catalytic activity: one cysteine residue, which is present in thiolate form, acts as a base and accepts a H^+ ion from L-DAP, whereas the other cysteine residue present in thiol form acts as an acid donate a H^+ ion to the molecule to form meso-DAP.[12] These two cysteine residues also form disulphide linkage under oxidized condition, which induces conformational changes at the catalytic site.[13] Previous structural studies of DAP epimerase suggest that the enzyme is monomeric in nature. But recent studies of DAP epimerase from *Escherichia coli* reveal that this enzyme is dimer in solution as well as in crystal state and dimerization is essential for optimal activity of this enzyme. Dimer interface occurs between the N-terminal domain of the two monomer subunits and the active site is located in a cleft between the two domains.[6,16] Superimposition of the catalytic sites of DAP epimerase structure from *Bacillus anthracis* and *Escherichia coli* demonstrated that the majority of active site residues are similarly oriented. In this study, we identified *E. faecalis* diaminopimelate epimerase (*Ef*DapF) as a drug target as it is absent in humans and is essential for the survival of bacterial species. Our study offers a new approach to aid in the design of inhibitors as novel antimicrobial agents. In this regards, we report here the cloning, over-expression and purification of *Ef*DapF. We also report structure prediction of *Ef*DapF, molecular docking studies and also binding studies using Fluorescence spectroscopic studies of *Ef*DapF with acetaminophen and dexamethasone.

2 | MATERIALS AND METHODS

2.1 | PCR amplification and purification of *Ef*DapF gene

The DAP-F gene encoding diaminopimelate epimerase was obtained by PCR amplification of the gene from genomic DNA of *Enterococcus faecalis* using forward and reverse primers (Dapf: 5′-ggaattcCATATGatgaatgttatgatgcaaaaagtac-3′) and (Dapr: 5′-ccgCTCGAGttatttttgagaaagtgtttgcaga-3′) containing *Nde*I and *Xho*I restriction sites, respectively. A PCR reaction was set up using 0.5U Phusion DNA polymerase (Thermo, California), 5X Phusion HF buffer, 0.2 mM dNTP mix (Thermo, California), 0.5 μM forward and reverse primers and 40 ng genomic DNA. The PCR cycling conditions were as follows: initial denaturation at 98°C for 60 seconds, followed by 35 cycles of denaturation at 98°C for 10 seconds, annealing at 60°C for 30s and extension at 72°C for 30 seconds with final extension at 72°C for 5 minutes performed in Bio-rad T100 Thermal cycler (Bio-rad, California). PCR product was analysed electrophoretically by using 2% Agarose gel with 100 bp DNA standard as reference. Furthermore, the amplified PCR product was purified with Wizard PCR and gel clean-up system kit as per manufacturer's instructions (Promega, Madison, Wisconsin).

2.2 | Cloning, over-expression and purification of *Ef*DapF protein

The PCR amplified *Ef*DapF gene was ligated with *Xho*I and *Nde*I digested pET28a vector using T4 DNA ligase (Thermo, California). The ligation reaction was set up at 4°C for overnight. The recombinant plasmid (pET28a + *Ef*DapF = pDapF) was transferred into the competent *E. coli* DH5α cells. The recombinant clone was confirmed for the insert by restriction digestion. Sequence of the cloned gene was confirmed by gene sequencing (Xcelris Labs Ltd., Ahmedabad, India). The recombinant plasmid was further transferred into the competent *E. coli* BL21 (DE3) cells for over-expression studies. A single BL21 (DE3) transformant from the transformed plate containing Pet28a + *Ef*DapF was inoculated in Luria-Bertani (LB) broth (Hi-media, Mumbai, India) containing 50 μg/mL kanamycin (Hi-media, Mumbai, India) and incubated at 37°C in shaker incubator overnight, then primary inoculation (1%) was added in LB broth (1000 mL) containing 50 μg/mL kanamycin and grown at 37°C to an optical density 600 of 0.6. Further 0.4 mM Isopropyl β-D-1-thiogalactopyranoside (IPTG) was added to the secondary culture for induction at 20°C overnight in shaker incubator. Furthermore, induced cells were harvested by centrifugation for 15 minutes at 8000*g* and lysed with 18 mL B-PER-II bacterial protein extraction reagent (Thermo, California). The cell debris was removed by centrifugation for 15 minutes at 11 000*g* at 4°C. The supernatant thus obtained was loaded onto a Ni-NTA Agarose resin (Qaigen, Hilden, Germany) column pre-equilibrated with buffer-A consisting of 150 mM NaCl, 50 mM Tris. Initial washing was done with 50 mL buffer-A, accompanied by washing with 50 mL buffer-B consisting of buffer-A and 30 mM Imidazole. The protein was eluted with 15 mL buffer-C consisting of buffer-A and 300 mM Imidazole, which was concentrated using Protein Concentrators PES, 10K MWCO (Thermo, California). Furthermore, the pooled fractions were dialysed by using Desalting Column (Thermo, California) pre-equilibrated with 20 mL buffer-A to remove lower molecular-weight contaminants and the protein sample was then further concentrated using centricon (Thermo, California) and stored at −20°C for further analysis.

2.3 | Molecular modelling of *Ef*DapF

Our initial search for three-dimensional (3D) structure of *Ef*DapF in the protein Data Bank was unsuccessful. Therefore, we started the procedure of developing a 3D model of DAP of *E. faecalis*. However, a BLAST search of the *Ef*DapF sequence against the PDB with default parameters was done to find templates for homology modelling of *Ef*DapF. Based on the maximum sequence identity, lower *E*-value, query coverage, the crystal structure of *Escherichia coli*, with sequence identity of 27% was selected as a template. The homology model of *Ef*DapF was developed in correlation with PDB ID 4IK0 as a template using SWISS-MODEL workplace (http://swissmodel.expasy.org).[17]

2.4 | Assessment and validation of 3D model

After modelling, the quality of the theoretically modelled protein structure was evaluated using Ramachandran plot and Protein Structure Analysis (ProSA) analysis with respect to stereochemical geometry and energy. PROCHECK analysis in PDBSUM platform of the model was done to calculate the Ramachandran plot[18] and ProSA analysis was carried out to evaluate energy and structural error for both template and modelled structure.[19,20]Finally, Discovery Studio Visualizer software (DassaultSystemes BIOVIA, Discovery Studio Visualizer, Release 2017, San Diego: DassaultSystemes, 2016) was used to visualize the structure and Pymol was used to calculate the structural difference between template and modelled structure.

2.5 | Docking studies with ligands

Docking studies were performed in order to estimate the binding interactions of potential inhibitors (acetaminophen and dexamethasone) with *Ef*DapF, using the AutodockVina Program.[21] The 3D structure of the potential ligand of acetaminophen and dexamethasone were obtained from the PubChem database (http://pubchem.ncbi.nlm.nih.gov) and were energy minimized by Autodock tools after adding Kollman charges and hydrogen atoms to the modelled structure. After preparing the ligands and protein, the grid box having dimensions $100 \times 106 \times 126$ Å in the coordinate XYZ with grid spacing of 0.375 Å were generated. To estimate the interaction energy and binding affinities between the docked ligand and the receptor protein, the Boryden-Fletcher-Goldfarb-Shanno-based AutoDockVina tool was used. Finally, the docked *Ef*DapF-Ligand complexes were evaluated on the basis of binding affinity (Kcal/mol) and RMSD (root-mean square deviation).

2.6 | Binding studies using fluorescence spectroscopy

The interactions of *Ef*DapF with ligands acetaminophen and dexamethasone were investigated using fluorescence spectroscopy (LS55, PerkinElmer, Waltham, Massachusetts). Scanning parameters for all the measurements were optimized with slit width of 9 nm for excitation and emission and scanned at a speed of 200 nm/min. The protein was excited at a wavelength of 280 nm and emission spectra were recorded within the range of 300 to 550 nm at 25°C. Quartz cell of 10 mm path length with 1 mL sample volume was used for emission measurements. The protein concentration was kept constant at 1×10^{-8} M and the concentration of acetaminophen and dexamethasone were varied as 10 μL, 20 μL, 30 μL and 40 μL, which were prepared from 1×10^{-7} M stock solution and the final ligand concentrations were 1 nM, 2 nM, 3 nM and 4 nM respectively. Spectral data of *Ef*DapF thus obtained with various concentrations of ligands were noted and plotted. The least square fit intensity changes for *Ef*DapF-acetaminophen and *Ef*DapF-dexamethasone binding curve were obtained using Sigma Plot 14.0. Each experiment was performed thrice and its average value was used for data analysis.[22-24]

3 | RESULTS

3.1 | Cloning, over-expression and purification of *Ef*DapF protein

Amplified *Ef*DapF gene was cloned, overexpressed and purified using affinity chromatography. The purity of protein was analysed using SDS-PAGE, which showed a single band around 36 kDa (Figure 1).

3.2 | In silico modelling and validation

Despite the importance of *Ef*DapF as a drug target, the 3D structure had not been reported prior to this study. Thus, it was a necessity to find the structure of *Ef*DapF to study its structural and biochemical relations. In this study, we determined the homology model of *Ef*DapF using in silico method. The *Ef*DapF gene encodes 326 amino acid residues long protein. Dap of *Escherichia coli* with PDB ID 4IK0 was selected as a template on the basis of maximum similarity (27%), QMEAN score of −4.18 and GMQE score of 0.57, which significantly validate the quality of the model. Figure 2 represents the overall representation of *Ef*DapF.

3.3 | Docking using ligands

The binding of acetaminophen and dexamethasone was found to be −8.3 and −5.3 kcal/mol, respectively. Docking studies revealed that acetaminophen was found to make hydrogen bonds with Asn72 and Asn13 (Figure 3) while dexamethasone interacted by forming hydrogen bonds with Asn205 and Glu223 (Figure 4) around the binding site. The docking of ligand with modelled *Ef*DapF is shown in Figure 5 (acetaminophen binding with *Ef*DapF) and Figure 6 (dexamethasone binding with *Ef*DapF).

3.4 | Binding studies using fluorescence spectroscopy

To validate the binding studies of *Ef*DapF with acetaminophen and dexamethasone, fluorescence spectroscopy techniques were done.

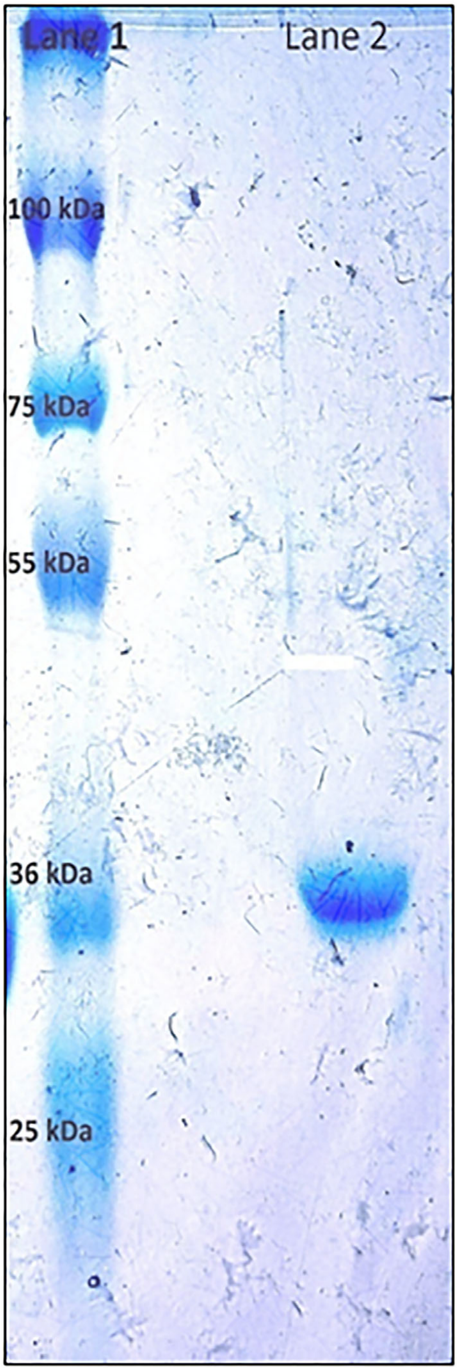

FIGURE 1 SDS-polyacrylamide gel electrophoresis showing purified *Ef*DapF protein at around 36 kDa region in Lane 1 against the protein marker in Lane 2

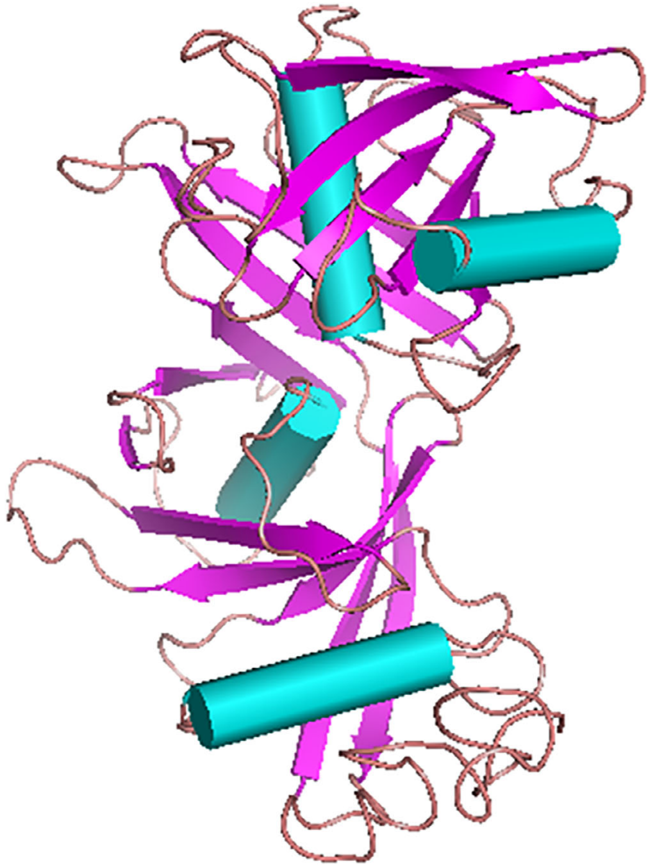

FIGURE 2 Overall representation of modelled *Ef*DapF protein

Protein concentration was kept constant while the concentrations of ligands were increased and binding affinity of both ligands was determined. It was observed that both acetaminophen and dexamethasone quenched the intrinsic fluorescence intensity of protein at 337 nm, which revealed that the compounds bound to *Ef*DapF. The fluorescence quenching coefficient (*Q*) was calculated using the following formula:

$$Q = (F_O - F)/F_O$$

where F_O represents the fluorescence in the absence of ligands and *F* represents the measured fluorescence. The *Q*% values thus obtained were plotted against the concentrations of ligands. The R^2 values, which provide an index of goodness of fit of the curve, were obtained using Sigma Plot 14.0 and were determined as .992 and .988 for acetaminophen and dexamethasone, respectively (Figure 7). The K_d values (equilibrium constant) for the binding of acetaminophen and dexamethasone were observed to be 2.69×10^{-8} M and 4.13×10^{-8} M, respectively.

4 | DISCUSSION

In *E. faecalis*, diaminopimelate epimerase (DapF) is a key enzyme for the biosynthesis of meso-DAP and lysine in the lysine biosynthetic pathway.[4-6] Meso-DAP and lysine, the products of this pathway, are involved in the synthesis of peptidoglycan, house-keeping proteins, virulence factors and thus survival of bacteria. In this regard, *Ef*DapF is considered as an important drug target against *E. faecalis*. Therefore, the design of tight inhibitors of *Ef*DapF may lead to new class of antibacterial drugs.[7] Also, because humans lack this enzyme, consequently *Ef*DapF represents a potential target for the development of anti-infective compounds.[8,9] We have identified the potential lead compounds (acetaminophen and dexamethasone) based on its

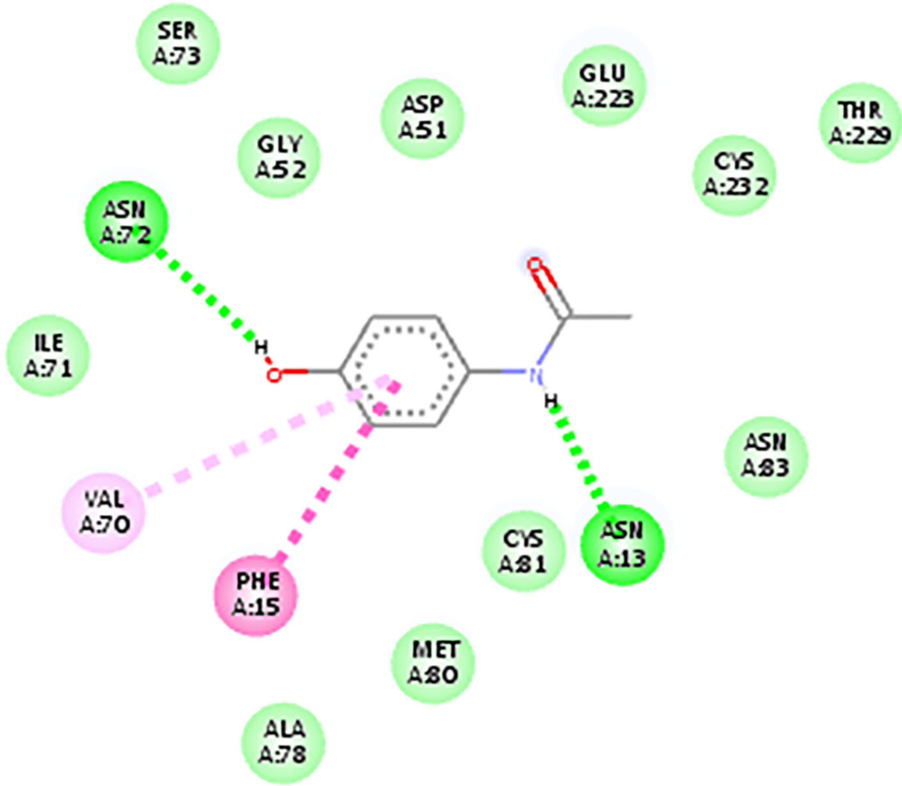

FIGURE 3 Docked structure of acetaminophen with the *Ef*DapF protein

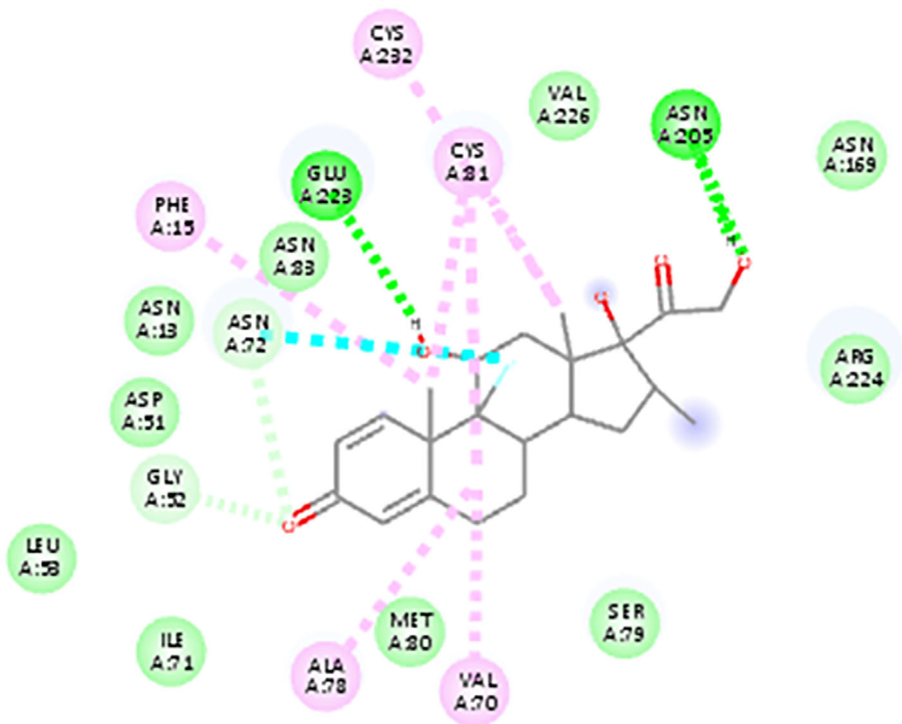

FIGURE 4 Docked structure of dexamethasone with the *Ef*DapF protein

structural similarity (homologous) with the known inhibitors of DapF. 3-Chlorodiaminopimelate[25] and 2-(4-amino-4-carboxybutyl)-2-aziridinecarboxylic acid[15,26] are the potent inhibitors of DapF and share slight structure similarity with acetaminophen and dexamethasone, respectively, and showed the inhibition of meso-Dap required for the synthesis of peptidoglycan, by blocking the activity of the enzyme. Due to its structural similarity with known inhibitors of DapF, we initially wanted to explore the binding of lead molecules to the enzyme and the site and nature of interactions in the binding site. In order to solve the 3D-structure of *Ef*DapF, in silico docking studies were done.[27-29] The

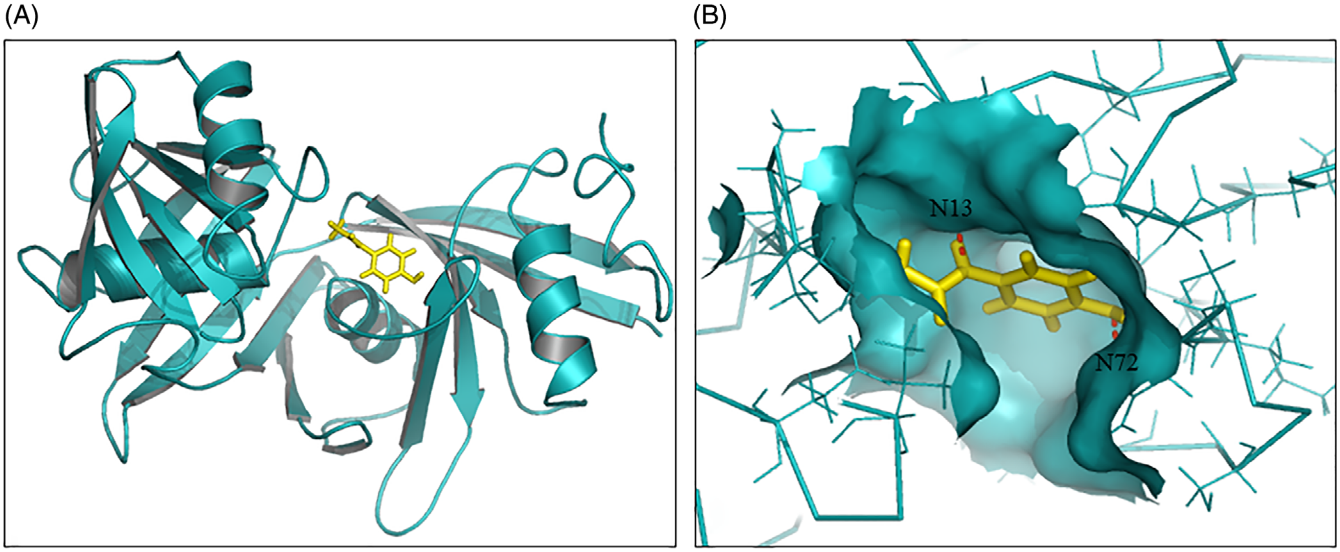

FIGURE 5 Acetaminophen binding in *Ef*DapF: A, Molecular surface diagram with ligand binding site, B, A close-up view of the ligand binding site of *Ef*DapF with the docked lead molecule showing hydrogen bonds with Asn72 and Asn13

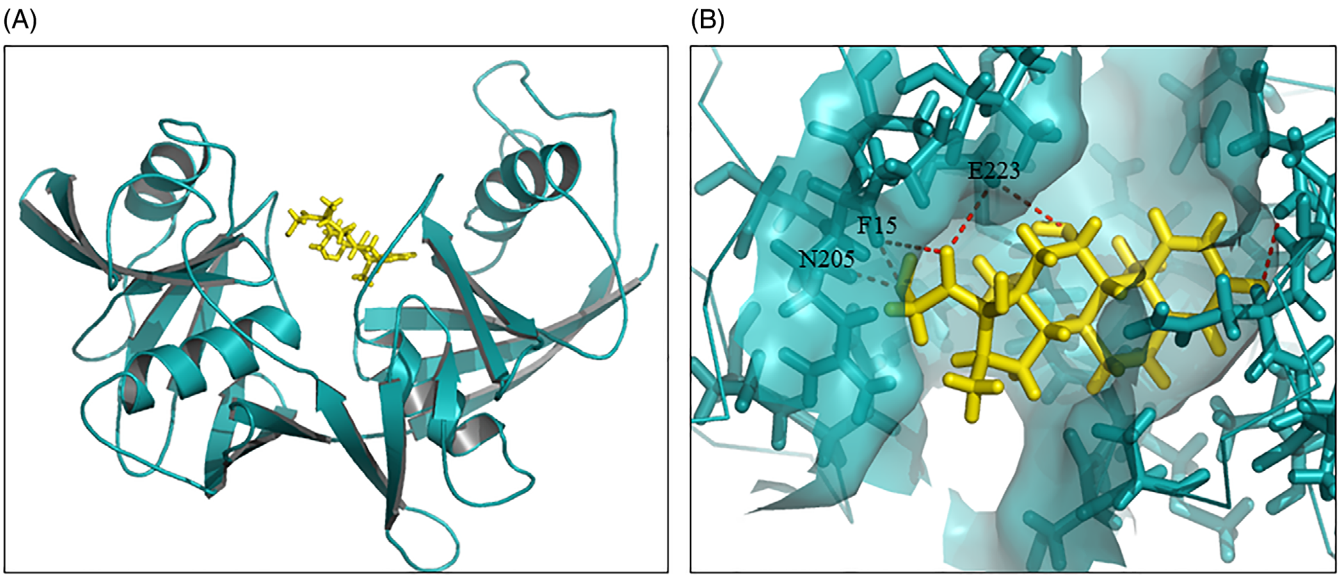

FIGURE 6 Dexamethasone binding in *Ef*DapF: A, Molecular surface diagram with ligand binding site, B, Magnified view of the protein-ligand interaction site, showing hydrogen bonds with Asn205 and Glu223

homology model of Dap was produced using the coordinates of crystal structure of *Escherichia coli* with PDB ID: 4IK0, the model showed maximum sequence similarity (Figure S1) of (27%), QMEAN score of −4.18 and GMQE score of 0.57, which significantly indicate the quality of the model. Docking studies showed a significant binding of acetaminophen and dexamethasone with *Ef*DapF with binding energy of −8.3 kcal/mol and −5.3 kcal/mol, respectively. *Ef*DapF-ligand complex was stabilized by forming various molecular interactions. Docking studies illustrated that acetaminophen was found to make hydrogen bonds with Asn72 and Asn13 (Figure 3) while dexamethasone interacted by forming hydrogen bonds with Asn205 and Glu223 (Figure 4) around the binding pocket. Apart from this, spectrofluorimetric binding analysis was used to study protein-ligand interactions. The K_d values (equilibrium constant) for the binding of acetaminophen and dexamethasone is 2.69×10^{-8} M and 4.13×10^{-8} M respectively, which is a good indication of strong protein-ligand interaction. Both fluorescence and binding studies confirmed the results obtained by in silico analysis. Furthermore, ligand

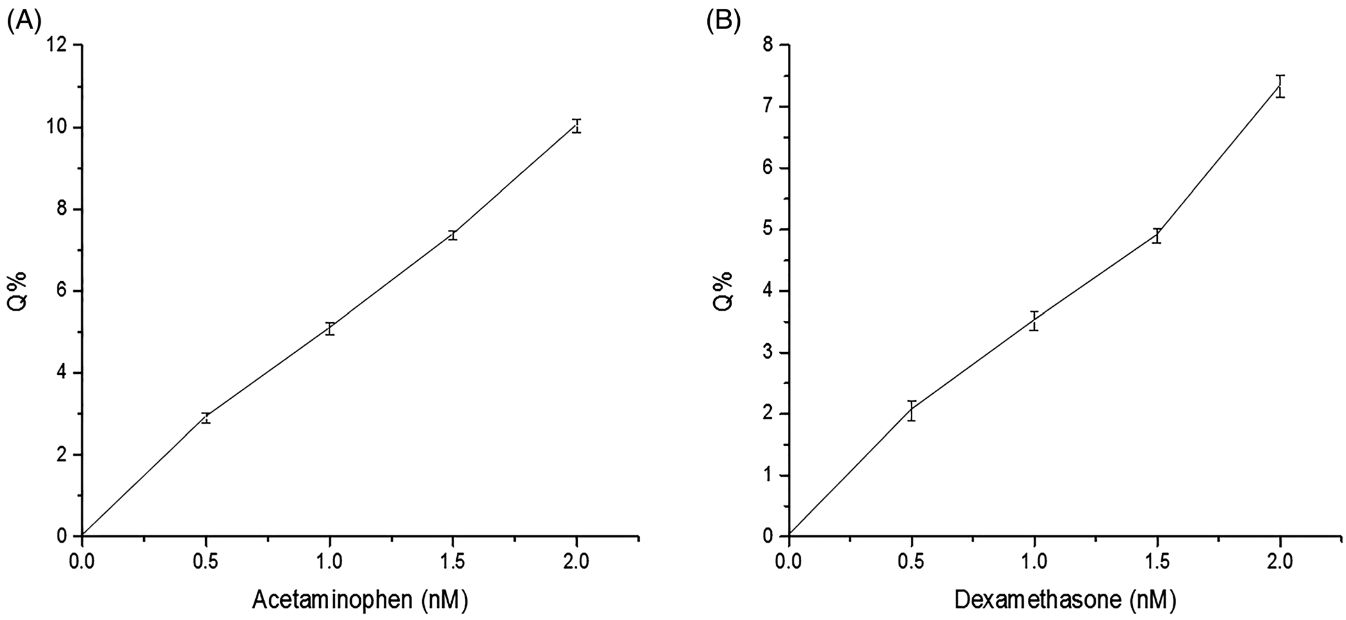

FIGURE 7 Binding curves show the bindings of ligands to *Ef*DapF and exhibit increase in fluorescence quenching as the concentration of the ligands increases from 10 to 40 nM. A, binding curve of *Ef*DapF-acetaminophen, B, binding curve of *Ef*DapF-dexamethasone

acetaminophen[30-33] and dexamethasone[34] are medicinal compounds and have antimicrobial activity. These are used either alone or in combination with other antibiotics. Thus, this study reports acetaminophen and dexamethasone as potential inhibitors of *Ef*DapF, which can be further explored and manipulated for developing tight inhibitors, which can be effective for the treatment of infections caused by MDR bacteria *E. faecalis*.

5 | CONCLUSION

This research was planned to study the binding of lead compounds (acetaminophen and dexamethasone) with *Ef*DapF and to further analyse the types of interactions and the binding pocket of the ligands on *Ef*DapF. Diaminopimelate epimerase (DapF) is a key enzyme for the biosynthesis of meso-DAP and lysine in the lysine biosynthetic pathway. Lysine is an essential amino acid in human beings, hence inhibition of bacterial lysine biosynthesis will not have any effect on human beings. Therefore, we have used computer-aided approach to analyse the interaction between protein and ligands, and fluorescence spectroscopy technique was used to conform the binding of ligands to *Ef*DapF. This study confirmed that these molecules bind to *Ef*DapF with significant interactions. Thus, these compounds can be further explored for developing tight inhibitors, which can be effective to treat the infection caused by MDR *E. faecalis*.

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CONFLICT OF INTEREST

None declared.

DATA AVAILABILITY STATEMENT

The authors did not have any special access privileges that others would not have. Others will be able to access these data in the same manner as the authors.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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